Microbiology

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[Page 1]
BASICS OF BIOSENSORS AND
MOLECULAR MOTORS
Unit 4
 Biosensors
 Types of Biosensors
 Components of biosensors
 Medical applications of biosensors
 Linear motors: actin and myosin
 Rotatory motors: flagella motor and ATPase

[Page 2]
What is a Biosensor?

[Page 3]
“Biosensor” – Any device that uses specific biochemical
reactions to detect chemical compounds in biological samples.
238 aa

[Page 4]
• 1916 First report on immobilization of
proteins : adsorption of invertase on activated
charcoal
• 1922 First glass pH electrode
• 1956 Clark published his definitive paper on
the oxygen electrode.
• 1962 First description of a biosensor: an
amperometric enzyme electrodre for glucose
(Clark)
• 1969 Guilbault and Montalvo –First
potentiometric biosensor:urease immobilized
on an ammonia electrode to detect urea
History of Biosensors

[Page 5]
• 1987 Blood-glucose biosensor launched by
MediSense ExacTech
• 1990 SPR based biosensor by Pharmacia
BIACore
• 1992 Hand held blood biosensor by i-STAT
• 1996 Launching of Glucocard
• 1998 Blood glucose biosensor launch by LifeScan
FastTake
• 1998 Roche Diagnostics by Merger of Roche and
Boehringer mannheim
• Current Quantom dots, nanoparicles, nanowire,
nanotube, etc
History of Biosensors

[Page 6]
• 1970 Bergveld –ion selective Field Effect Transistor (ISFET)
• 1975 Lubbers and Opitz described a fibre-optic sensor with
immobilised indicator to measure carbon dioxide or oxygen.
• 1975 First commercial biosensor (Yellow springs
Instruments glucose biosensor)
• 1975 First microbe based biosensor, First immunosensor
• 1976 First bedside artificial pancreas (Miles)
• 1980 First fibre optic pH sensor for in vivo blood gases
(Peterson)
• 1982 First fibre optic-based biosensor for glucose
• 1983 First surface plasmon resonance (SPR)
immunosensor
• 1984 First mediated amperometric biosensor:
ferrocene used with glucose oxidase for glucose detection
History of Biosensors

[Page 7]
Current Definition
A sensor that integrates a biological element with a physiochemical
transducer to produce an electronic signal proportional to a single analyte
which is then conveyed to a detector.

[Page 8]
1. LINEARITY Linearity of the sensor should be high
forthe detection of high substrate
concentration.
2. SENSITIVITY Value of the electrode response per

substrate concentration.
3. SELECTIVITY Chemicals Interference must be
minimised for obtaining the correct
result.
4.RESPONSE TIME Time necessary for having 95%
of the response.
Basic Characteristics of a
Biosensor

[Page 9]
1. The Analyte A substance of interest that needs detection. For instance, glucose is an ‘analyte’ in a biosensor designed to detect glucose
Protein, toxin, peptide, vitamin, sugar, metal ion.
2. Sample handling Micro fluidics - Concentration increase/decrease,
Filtration/selection
Components of Biosensor
3. Bioreceptor A molecule that specifically recognises the analyte is known as a bioreceptor. Enzymes, cells, aptamers, deoxyribonucleic acid (DNA) and antibodies are some examples of bioreceptors. The process of signal generation (in the form of light, heat, pH, charge or mass change, etc.) upon interaction of the bioreceptor with the analyte is termed bio-recognition.
4. Transducer The transducer is an element that converts one form of energy into another. In a biosensor the role of the transducer is to convert the bio-recognition event into a measurable signal. Most transducers produce either optical or electrical signals that are usually proportional to the amount of analyte–bioreceptor interactions.
5. Detection/Recognition How do you specifically recognize the analyte?
6. Signal (How do you know there was a detection)

[Page 10]
Components of a Biosensor
Detector

[Page 11]
Example of biosensors
Pregnancy test
Detects the hCG protein in
urine.
Glucose monitoring device (for diabetes patients)
Monitors the glucose level in the blood.

[Page 13]
Example of biosensors
Infectious disease

biosensor from RBS
Old time coal miners'
biosensor

[Page 14]
Research Biosensors
Biacore Biosensor platform

[Page 15]
✓Fluorescence
✓DNA Microarray
✓SPR Surface plasmon resonance
✓Impedance spectroscopy
✓SPM (Scanning probe microscopy, AFM, STM)
✓QCM (Quartz crystal microbalance)
✓SERS (Surface Enhanced Raman Spectroscopy)
✓Electrochemical
Typical Sensing Techniques
for Biosensors

[Page 16]
Types of Biosensors
1. Calorimetric Biosensor
2. Potentiometric Biosensor
3. Amperometric Biosensor
4. Optical Biosensor
5. Piezo-electric Biosensor

[Page 17]
Calorimetric Biosensors
If the enzyme catalyzed reaction is exothermic, two thermistors may be used to measure the difference in resistance between reactant and product and, hence, the analyte concentration.

[Page 18]
Potentiometric Biosensor
– For voltage: Change in distribution of charge is detected using ion-selective electrodes, such as pH-meters.

[Page 19]
Piezo-Electric Biosensors

The change in frequency is proportional to
the mass of absorbed material.
Piezo-electric devices use gold to detect the specific
angle at which electron waves are emitted when the
substance is exposed to laser light or crystals, such as
quartz, which vibrate under the influence of an electric
field.

[Page 20]
Electrochemical Biosensors
• For applied current: Movement of e- in redox
reactions detected when a potential is applied
between two electrodes.

[Page 21]
Optical Biosensors
•Colorimetric for color
Measure change in light adsorption
•Photometric for light intensity
Photon output for a luminescent or
fluorescent process can be detected with
photomultiplier tubes or photodiode
systems.

[Page 22]
Motivated by the application to clinical diagnosis and
genome mutation detection
Types DNA Biosensors
• Electrodes
• Chips
• Crystals
DNA biosensor

[Page 23]
Electrochemical DNA Biosensor
▪ Steps involved in electrochemical
DNA hybridization biosensors:
▪ Formation of the DNA recognition layer
▪ Actual hybridization event
▪ Transformation of the hybridization event
into an electrical signal

[Page 24]
Wearable Biosensors
Ring Sensor

Smart Shirt

[Page 25]
Biosensors on the Nanoscale
❑ Molecular sheaths around the nanotube are developed that respond to a particular chemical and modulate the nanotube's optical properties.
❑ A layer of olfactory proteins on a nanoelectrode react with low-concentration odorants (SPOT-NOSED Project). Doctors can use to diagnose diseases at earlier stages.
❑ Nanosphere lithography (NSL) derived triangular Ag nanoparticles are used to detect streptavidin down to one picomolar concentrations.
❑The School of Biomedical Engineering has developed an anti- body based piezoelectric nanobiosensor to be used for anthrax,HIV hepatitis detection.

[Page 26]
❖ Food Analysis
❖ Study of biomolecules and their interaction
❖ Drug Development
❖ Crime detection
❖ Medical diagnosis (both clinical and laboratory use)
❖ Environmental field monitoring
❖ Quality control
❖ Industrial Process Control
❖ Detection systems for biological warfare agents
❖ Manufacturing of pharmaceuticals and replacement organs
Application of Biosensor

[Page 27]
Potential Applications
• Clinical diagnostics
• Food and agricultural processes
• Environmental (air, soil, and water) monitoring
• Detection of warfare agents.

[Page 28]
• Biosensors play a part in the field of environmental quality, medicine and industry mainly by identifying material and the degree of concentration present

[Page 29]
▪Molecular Machines/Motors
▪Properties of ATP-based Protein Molecular
Machines
▪The F0F1-ATP Synthase Motors
▪Coupling and Coordination of Motors
▪The Bacterial Flagellar Motor
▪Flagellar motor structure

[Page 30]
Molecular Machines/Motors
• Molecular machines can be defined as devices that
can produce useful work through the interaction of
individual molecules at the molecular scale of length.
• A convenient unit of measurement at the molecular
scale would be a nanometer. Hence, molecular
machines also fall into the category of nanomachines.
• As our knowledge and understanding of these
numerous machines continues to increase, we now
see a possibility of using the natural machines, or
creating synthetic ones from scratch, by mimicking
nature.

[Page 31]
Different types of molecular machines. These machines used by nature
for force generation and motion. They convert chemical energy into
mechanical force via conformational changes.

[Page 32]
• Most of the natural machinery is built from
proteins.
• These findings help unravel the mysteries
associated with the molecular machinery and
pave the way for the production and
application of these miniature machines in
various fields, including medicine, space
exploration, electronics and military.
• We divide the molecular machines into three
broad categories—protein based, DNA-
based, and chemical molecular motors.

[Page 33]
Properties of ATP-based Protein Molecular Machines

• Two form the F0F1-ATP synthase, and the third one is the bacterial flagellar motor.
• The protein-based molecular motors rely on an energy-rich molecule known as adenosine triphosphate (ATP),
• The machines described in this section, the F0F1-ATPase, the kinesin, myosin, and dynein superfamily of protein molecular machines, and bacteria flagellar motors all depend, directly or indirectly, on ATP for their input energy.
• These machines have now been segregated out of their natural environment and are seen as energy conversion devices to obtain forces, torques, and motion.
• One disadvantage associated with ATP dependence is that the ATP creation machinery itself could be many times heavier and bulkier than the motors, thereby making it more complex.
• These machines perform best in their natural environment, and in the near future it may be possible to have them as a part of biomimetic molecular machinery.

[Page 34]
The F0F1-ATP Synthase Motors
• ATP is regarded as the energy currency of biological systems.
• When this currency is utilized (i.e., the energy of the molecule that is used to drive a biological process), the terminal anhydride bond in the ATP molecule has to be split. This leaves adenosine diphosphate (ADP) and a phosphate ion (Pi) as the products, which are recombined to form ATP by a super efficient enzyme motor assembly called the F0F1-ATP synthase (F0F1-ATPase).
• ATP synthase is present inside the mitochondria of animal cells, in plant chloroplasts, in bacteria, and some other organisms.

[Page 35]
• ATP synthase consists of two portions: a membrane-spanning portion, Fo, comprising the ion channel, and a soluble portion, F1, containing three catalytic sites.
• Both Fo and F1 are reversible rotary motors --- perhaps the smallest motors known to science. Fo uses the transmembrane electrochemical gradient to generate a rotary torque to drive ATP synthesis in F1 or, when driven backwards by the torque generated in F1, to pump ions uphill against their transmembrane electrochemical gradient.

• F1 generates a rotary torque by hydrolyzing ATP at
its three catalytic sites or, when turned backwards by
the torque generated in Fo, as a synthesizer of ATP.

[Page 37]
The F0F1-ATPase motors. The F0 motor is embedded in the inner mitochondrial
membrane of the mitochondria. F0 is typically composed of a, b, and c subunits as
shown. The F1 motor is the soluble region composed of three α-, three β-, one each
of γ -, δ- and ε-subunits.

[Page 38]
Coupling and Coordination of Motors
• The ATP synthase is actually a combination of two motors
functioning together, the hydrophobic transmembrane F0-
ATPase motor and the globular F1-ATPase motor
The F0F1-ATPase contains two rotary motors: the membrane –bound
F0, driven by the flux of ions across a membrane, and the soluble F1,
driven by ATP hydrolysis. These two motors are coupled by sharing
a common motor, drawn in white, and a common stator, drawn dark.
The figure shows the F0F1-ATPase operating as an ATP-synthase.
Rotation of F0 driven by ion flux drives F1 in reverse, causing it to
synthesize ATP from ADP and inorganic phosphate.

[Page 39]
The Bacterial Flagellar Motor
• A flagellum (plural: flagella) is a
long, slender projection from
the cell body, composed of
microtubules and surrounded
by the plasma membrane.
• In small, single-cell organisms
they may function to propel the
cell by beating in a whip-like
motion; in larger animals, they
often serve to move fluids along
mucous membranes such as the
lining of the trachea.
• The bacterial flagellar motor is
a nanotechnological marvel, no
more than 50 nm in diameter,
built from about 20 different
kinds of parts.
Examples of bacterial flagella
arrangement schemes. A-

Monotrichous; B-Lophotrichous;
C-Amphitrichous; D-Peritrichous

[Page 40]

• It spins clockwise (CW) or counterclockwise (CCW) at speeds on the order of 100 Hz, driving long thin helical filaments that enable cells to swim.
• Peritrichously flagellated cells (peri, around; trichos, hair), such as Escherichia coli, execute a random search, moving steadily at about 30 diameters per second, now in one direction, now in another.
• Steady motion requires CCW rotation. Receptors near the surface of the cell count molecules of interest (sugars, amino acids, dipeptides) and control the direction of flagellar rotation.
• If a leg of the search is deemed favorable, it is extended, i.e., the motors spin CCW longer than they otherwise would. This bias enables cells to actively find regions in their environment where life is better.

[Page 41]

• Thus, the flagellar motor is the output organelle of a remarkable sensory system, the components of which have been honed to perfection by billions of years of evolution.
• A number of bacterial species in addition to E. coli depend on flagella motors for motility: e.g., Salmonella enterica serovar, Typhimurium (Salmonella), Streptococcus, Vibrio spp., Caulobacter, Leptospira, Aquaspirrilum serpens, and Bacillus.
• The rotation of flagella motors is stimulated by a flow of ions through them, which is a result of a build-up of a transmembrane ion gradient. There is no direct ATP-involvement; however, the proton gradient needed for the functioning of flagella motors can be produced by ATPase.

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Flagellar motor structure
• Bacterial flagella are the only biological structures known that use rotation for the purpose of locomotion.
• Flagella consist of a rotary motor embedded in the cell envelope connected to an extracellular helical propeller. The motor is powered by the flow of ions down an electrochemical gradient across the cytoplasmic membrane into the cell.
• The ions are typically H+ (protons), although certain marine and alkalophilic species have motors driven by Na+.
• The electrochemical gradient (protonmotive force or sodium motive force) consists of a transmembrane voltage and a concentration difference across the membrane, both of which are maintained by various metabolic processes.
• The rotor shown in white (Figure 5), consists of a series of rings spanning the cell envelope and is attached via the flexible hook to the helical propeller or filament. The stator is a ring of particles in the cytoplasmic membrane, containing the proteins MotA and MotB, and anchored to the peptidoglycan cell
cell

[Page 43]
The bacterial flagellar motor. The rotor, drawn in white, consists of a series of rings that span
the cell envelope and are attaches to the extracellular hook and filament. The stator consists
of a ring of torque-generating units containing the proteins MotA and MotB and anchored to the cell wall. Ions flowing through the motor generate torque by means of unknown interactions between the rotor and stator in the vicinity of the C-ring. The F0F1-ATPase is also shown, to the same scale.

[Page 44]
The fuel/air intake system red, the electrical system green, and the exhaust system blue

[Page 45]
Cytoskeleton: Microtubules, Microfilaments, and Intermediate filaments
• Cytoskeleton gives the cell shape, structure, and physical organization
• Motor proteins can rearrange the structural elements, or move cell components around the cytoskeleton.
• The cytoskeleton is in constant state of change depending on the

requirements of the cell.
• 3 major structural elements of the
cytoskeleton
• Microtubules - hollow, rigid
cylindrical tubes made from
tubulin subunits
• Microfilaments - solid, thinner
structures made of actin
• Intermediate filaments - tough,
ropelike fibers made of a variety of
related proteins
Mitochondrion
Plasma membrane
Ribosomes
Rough
Endoplasmic
reticulum

[Page 46]
The Myosin Linear Motor
• Myosin is a diverse superfamily of motor proteins.
• Myosin-based molecular machines transport cargoes along actin
filaments, the two-stranded helical polymers of the protein actin
that are about 5–9 nm in diameter. They do this by hydrolyzing
ATP and utilizing the released energy.
• In addition to transport, they are also involved in the process of
force generation during muscle contraction, wherein thin actin
filaments and thick myosin filaments slide past each other.
• Myosin molecules were first seen (in the late 1950s) through
electron microscope protruding out from thick filaments and
interacting with the thin actin filaments. Since that time, ATP
has been known to play a role in myosin-related muscle
movement along actin; however, the exact mechanism was
unknown, until it was explained in 1971.

[Page 47]
Myosin cross-bridge model of contraction. Myosin uses the
energy released from ATP hydrolysis to move along actin.

[Page 48]
The Kinesin Linear Motor
• The kinesin and dynein families of proteins are involved in cellular
cargo transport along microtubules, in contrast to myosin, which
transports along actin.

• Microtubules are 25-nm diameter tubes made of protein tubulin and are present in the cells in an organized manner.
• Microtubules have polarity; one end being the plus (fast-growing) end while the other end is the minus (slow-growing) end.
• Kinesins move from the minus end to the plus end of the microtubule, whereas dyneins move from the plus end to the minus end.
• Microtubule arrangement varies in different cell systems. In nerve axons, they are arranged longitudinally such that their plus ends point away from the cell body and into the axon.
• In epithelial cells, their plus ends point toward the basement membrane.
• They extend radially out of the cell center in fibroblasts and macrophages with the plus end protruding outward.
• Similar to myosin, kinesin is also an ATP-driven motor.
• One unique characteristic of the kinesin family proteins is their processivity; they bind to microtubules and literally walk on it for many enzymatic cycles before detaching.

[Page 50]
The Dynein Motor
• The dynein superfamily of proteins was discovered in 1965.
• Dyneins exist in two isoforms: cytoplasmic and axonemal.
• Cytoplasmic dyneins are involved in cargo movement, whereas axonemal dyneins are involved in producing bending motions of cilia and flagella.
• Because dynein is a larger and more complex structure than other motor proteins, its mode of operation is not as well known. However, electron microscopy and image processing was used to show the structure of a flagellar dynein at the start and end of its power stroke, which gives some insight into its possible mode of force generation.

[Page 51]
Cilia and flagella have a core axoneme, a complex of microtubules and associated proteins.

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[Page 1]
Metals
Semiconductor

Materials
Ceramics
Polymers
Synthetic
BIOMATERIALS
Orthopedic
screws/fixation
Dental Implants Dental Implants
Heart
valves
Bone
replacements
Biosensors
Implantable
Microelectrodes
Skin/cartilageDrug Delivery
Devices Ocular
implants

[Page 2]
Biomaterial Science

[Page 3]
Bileaflet Heart Valves

[Page 4]
Dental Implants

[Page 5]
Intraocular Lenses

- Implantation often performed on outpatient basis.

[Page 6]
Vascular Grafts

- Must Be Flexible.
- Designed With Open Porous Structure.
- Often Recognized By

Body As Foreign.

[Page 7]
Hip
-
Replacements

[Page 8]
Normal versus Arthritic Hip
Normal Hip: note the space
between the femur and
acetabulum, due to cartilage
Arthritic Hip: No space
visible in joint, as
cartilage is missing

[Page 9]
using a Polymethylmethacrylate
(PMMA) cement to adhere the
metal to the bone
using a porous metal
surface to create a bone
ingrowth interface
Two attachment methods

[Page 10]
the acetabulum and the proximal
femur have been replaced. The
femoral side is completely metal.
The acetabular side is composed
of the polyethylene bearing
surface
Overview of femoral
replacement

[Page 11]
The two materials are
bonded and equal force is
applied to both
Load transfer in Composite materials

[Page 12]
Comparison: Moduli of Elasticity
Modulus of elasticity of different implant materials and bone (in GPa)

[Page 13]
Implant bonding
A bonded interface is
characteristic of a
cemented prosthesis
(left)
non-bonded interface is
characteristic of a non-
cemented press fit
prosthesis (right)

[Page 14]
Degradation Problems
Example of fractured
artificial cartilage from
a failed hip
replacement

[Page 15]

Chapter 4
Biomaterials: Basic principles
4.1 Introduction
Human tissues and organs sometimes fail to perform their regular actions due to genetic defects, age, illness, trauma, degeneration, or injuries. Some of these conditions are handled with the use of day-to-day medica - tion, i.e., drugs. However, some cannot be repaired/rectified by providing medicines and require the use of unique materials and devices. These then bring about the inevitability of surgical repair, which encompasses ana - tomical sections such as knee joints, elbow joints, vertebrae, teeth, and other crucial organs such as heart, skin, kidney, etc. In the broadest sense, the unique materials (other than drugs) or combinations of materials that are primarily expected to be used inside a mammal or human to treat, repair, augment or replace any tissue is referred to as biomaterials. These biomaterials can either be formed from nature or physically manufactured by utilizing a wide range of physical and chemical methods.
The field of biomaterials is multidisciplinary (Figure 4.1). The design of a simple biomaterial (for example, a bone screw or bone plate) neces -

sitates knowledge and ideas from multiple disciplines. It needs the syner -
gistic integration of materials, biology, medicine, mechanical sciences,
and chemistry. The number of medical devices used annually by humans
is significant, estimated at 1.5 million by the World Health Organization,
with around 10,000 forms of standardized device classes available
worldwide.
The term “biomaterials” is often described as any material that comes
into contact with humans or animals to fulfil their intended function without
causing any toxic reaction. This is the single most crucial aspect that dif -
ferentiates a biomaterial from any other material, i.e., its capacity to be in

contact with human body tissues without instigating an undesirable degree
of response. For thousands of years, humans have inevitably used or at least
attempted to use materials to make devices and instruments of practical
value by converting basic available substances into materials. Biomaterials
have a long history of medical use, and at various times they were seen in a
different way. Based on the development and its uses, the broad definition
has evolved over the years and could be further expanded as new applica-
tions in medicine emerge. Nevertheless, the meaning of biomaterials
attained a harmony of opinion by scientists worldwide and is now defined
now as “a material designed to take a form that can direct, through interac-
tions with living systems, the course of any therapeutic or diagnostic proce-
dure.” In most cases, a biomaterial is any natural or synthetic biocompatible
material that is used to replace or assist part of an organ or a tissue while
maintaining close contact with living tissue. They may be produced from
Figure 4.1: Interdisciplinary system of biomaterials (Cooke et al., 1996; Marin et al.,
2020; Ramakrishna et al., 2010; Ratner, 1996).

materials including solid, liquid, and gel substances (metallic components, polymers, ceramics or composite materials). It should be noted that the bio prefix of biomaterials applies to biocompatible, rather than biological or biomedical, as many would intuitively assume.

Biomaterials, a fascinating and highly interdisciplinary area, have grown to become an integral component in the modern-day improvement of human condition and quality of life. It is accomplished by addressing numerous health-related issues that come from several sources. Over the past few decades, biomaterials have broadened their applications from diagnostics (gene arrays and biosensors) and medical equipment (blood bags, surgical tools) to therapeutic medications (medical implants and devices) and emerging regenerative drugs (tissue-engineered skin and cartilage), and more. Many uses of synthetic and naturally occurring medicinal materials can be classified by their place and role in the human body: skeletal system (joint replacements, bone fractures and defects, artificial tendons and ligaments, dental implants), cardiovascular system (blood vessel prosthesis, heart valves), organs (artificial heart, skin repair), etc. The vital applications are presented in Figure 4.2 and tabulated in Table 4.1.

Figure 4.2: Biomaterials have made an enormous impact on the treatment of injury and disease and are used throughout the body (Cooke et al., 1996; Kuhn, 2005; Ratner et al., 2020).

Table 4.1: Applications of biomaterials.

Organ/Tissue	Examples
Heart	Biventricular pacemakers, artificial valves, artificial heart
Eye	Contact lens, artificial intraocular lens
Ear	Hair-clip artificial stapes, cochlear implants
Muscle	Sutures
Kidney	Dialysis machines
Skin	Skin wound repair, burn dressings, skin substitutes, artificial skin
Circulation	Synthetic blood vessels
Bone	Bone screws, plates, pins, intramedullary rods, bone cement to repair defects
Teeth	Fillings and replacements

4.2 Desired properties of biomaterials

As discussed earlier, the ultimate goal of biomaterials in medicine is to treat, improve, or substitute tissue organs (e.g., bone, muscle, skin) or body function. Such goals can be accomplished by integrating the properties of materials, nature of the system, device architecture, and physiological specifications. The biomaterial application method must combine the chemical and mechanical features of the biological system in order to achieve the required functional results (Figure 4.3). For the processing of biomaterials and related medical items, different elements from science and social parameters are also considered.

4.2.1 Biocompatibility

From a biological point of view, biocompatibility is nothing but the acceptability of non-living materials (synthetic or natural) in a living body (mammal/human). This acceptance is broadly defined as the " ability of a material to perform its desired functions with respect to a medical ther - apy, to induce an appropriate host response in a specific application and to interact with living systems without having any risk of injury, toxicity, or rejection by the immune system and undesirable or inappropriate local or systemic effects". Since a biomaterial is intended to be used in intimate contact with living tissues, it is important that the implanted material does

not cause any undesired reactions to the surrounding tissues and host organs. Apart from that, the implanted material is expected not to subdue the activity of normal cells or provoke any unwanted local or systemic reactions in the recipient, but is encouraged to generate positive cellular or tissue response.

Usually, the criteria for this biocompatibility are dynamic and detailed and differ with individual specific medical applications. For example, a specially designed material (screw or plate) may possibly be biocompat - ible in bone implant surgery, but the same components may not be bio - compatible in skin applications.

4.2.2 Host response

Host response is defined as the "response of the host organism (local and systemic) to the implanted material or device" . Most of the materials are never inert, and a biomaterial's clinical success depends significantly on the host tissue's reaction with the foreign material. For material establishment-host tissue interaction, these reactions depend heavily on the time-length, purpose, and site of implantation.

4.2.3 Non-toxicity

Toxicity of biomaterials deals with elements that migrate out of the bio -
materials. A carefully designed biomaterial should serve its purpose in the

Figure 4.3: Primary requirements of new material design.

[Page 20]

living body’s environment without negatively influencing other cells, organs, or the whole organism. It is logical to assume that, unless specifically designed to do so, a biomaterial should not release or generate anything from its bulk.

4.2.4 Mechanical properties

In addition to biocompatibility, mechanical properties are embedded in biomaterial design prior to implementation and will majorly contribute to the outcome. Materials undergo several forces which are primarily stress, strain, shear, and a mixture of these. The tensile strength, yield strength, elastic modulus, corrosion, creep, and hardness are some of the most important properties of biomaterials that should be carefully studied and evaluated before implantation. For hard tissue applications, the mechanical properties are of top priority.

4.2.5 Corrosion, wear, and fatigue properties

Wear is often considered as one of the leading causes of implant failure. In some cases, wear has also been shown to accelerate the corrosion of biomedical devices and implants. The fatigue resistance is related to the material’s reaction to repeated cyclic loads.

4.2.6 Design and manufacturability

Appropriate material design is also one of the critical factors to consider for biomaterials. For example, in the early 1900s, bone plates were utilized to assist in fixing long bone fractures. Many of these experimental plates broke because of their primitive mechanical design where they were too narrow and had a stress-focusing corner. Manufacturability is the ability to manufacture the item with relative ease that is ideal for its intended use at minimal cost and high reliability. Concerning biomaterials, the manufacturability, in a broader sense, incorporates the potential of the material to be sterilized by a validated sterilization technique which is deemed appropriate for biomedical applications. The material is

[Page 21]

expected to not get impaired by standard sterilizing techniques such as autoclaving, dry heat, radiation, ethylene oxide, etc. (see Chapter 8 for more details).

In summary, for any material to qualify as a biomaterial, it should

- Be biocompatible, i.e., non-toxic, non-carcinogenic, non-allergenic, etc.
- Have required/suitable physical and chemical properties
- Have suitable mechanical properties
- Have stable durability for the period it is intended for, i.e., hours to years
- Be easy to process with the available techniques
- Be sterilizable with current facilities without any difficulty
- Be cost-effective and accessible

4.3 Types of biomaterials

Depending on the chemical bonding, materials can be categorized into three broad groups. These are (i) polymers, (ii) ceramics, and (iii) metals. Because these material structures vary by their nature of bonding (cova - lent, ionic or metallic), they have different properties, and hence different uses in the body (Table 4.2). Different kinds of biomaterials and chemical bonds associated with it are shown in Figure 4.4. Metals and ceramics are differentiated in the field of materials engineering, although they are both

Figure 4.4: Different kinds of biomaterials and chemical bonds associated with them (Gul et al., 2020; He et al., 2017; Zindani et al., 2019).

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Table 4.2: Classification of biomaterials and typical properties associated with them (Wagner, 2020).

Attributes Polymers Metals & Alloys Ceramics

Type of bonds
present
Covalent & van der
Waals forces
Metallic Ionic/Covalent
Melting point Low Intermediate High
Chemical stability Poor Good Very high
Electrical
conductivity
Very low High Very low, but varies
Thermal
conductivity
Very low to
intermediate
High Low
Properties and
advantages
Degradable, inert,
similar density to
soft tissues and ease
of processing
High strength and
hardness
Non-conductive and
inert; closely
mimic biological
properties of bone
Mechanical
deformation
Very high, plastic (can
be easily shaped and
processed)
High (ductile) Low (brittle)
Major issues Thermally unstable;
low strength
Wear and
corrosion
High density and
brittle
Biomedical
applications
Soft tissue implants;
drug delivery
systems; tissue
engineering

Hard tissue
applications
(Orthopedic and
dental implants)
Tissue engineering

inorganic compounds. Each material has its own benefits and drawbacks, and its biomedical applications are decided according to its individual properties and intended place substitution.

4.3.1 Metals

Metals are the most widely known biomedical materials and are indis - pensable in the medical field. Nearly all of the metal biomaterials are crystalline in nature, i.e., with regular atomic arrangements. Metals have great strength, resistance to fracture toughness, better elasticity, and rigidity compared with ceramics and polymers. Their outstanding

mechanical reliability properties are controlled by dislocation and crystallization. For this reason, metals are extensively employed for load-bearing implant applications such as orthopaedic, dental, and maxillofa - cial surgery. Apart from these, metals are also used in making stents and stent-grafts for cardiovascular surgeries. The most common metals and alloys used for biomedical applications are stainless steel, titanium, titanium-based alloys, cobalt-based alloys, magnesium-based alloys, and tantalum-based alloys.

4.3.2 Ceramics

Ceramics are inorganic solid materials consisting of metallic and non-metallic elements that are predominantly bound together by ionic bonds. They exist as both crystalline and non-crystalline (amorphous) com - pounds. Ceramics are typically characterized by excellent biocompatibil-ity, high corrosion resistance, high wear, high strength, extremely high stiffness, and hardness. The advancement of ceramic material applications in the biomedical industry has focused mostly on orthopaedics and den - tistry. Bioinert ceramics such as alumina (Al_2O_3), zirconia (Zr_2O_3) and pyrolytic carbon, and bioresorbable ceramics such as calcium phosphates are some of the widely employed bioceramics.

4.3.3 Polymers

Polymers are macromolecules and they represent a major and versatile class of biomaterials being widely applied in biomedical applications due to their low toxicity in biological fluids, easy pre/post-processing, sterilization, better shelf life, lightweight nature, and remarkable physical and chemical properties.

Biomaterial use is primarily dependent on the need, function, and environment of the intended application. In a variety of implants, metal, polymers, ceramics, and composites are being used. The specific material or material combinations for use in a device will significantly affect both its patient performance and marketing potential. The next three chapters will discuss in detail each of the material types.

4.4 Multiple choice questions

1. Biomaterials
(a) are always synthetic materials; natural materials are not employed
(b) are always natural materials; synthetic materials are not employed
(c) can be natural or synthetic materials
(d) are always polymeric materials

2. Select the statement which correctly relates to biocompatibility.
(a) a biocompatible material should provide healing characters
(b) a biocompatible material should have therapeutic characteristics
(c) a material is considered as a biocompatible material as long as it causes no harm to the host body
(d) a biocompatible material should have the exact dimensions as the damaged tissue or part

3. Select the option(s) which do(es) NOT come under the class of biocompatible materials.
(a) eyeglasses; wheelchair
(b) contact lenses; dental implants
(c) orthopedic implant; stents
(d) external hip prosthesis; massage footwear

4. Which of the following has the best osteointegration properties?
(a) SS316
(b) porous titanium
(c) Co-Cr alloys
(d) all of the above

5. Which of the following is/are biomaterial(s)?

I. materials used for tooth filling
II. materials used for cardiovascular repairs
III. glucose meters and stethoscopes
IV. materials used for hip implants
(a) only I
(b) only I, II and IV
(c) only I, III and IV
(d) only III

6. Select the option(s) which is/are TRUE about biodegradation.
(a) it depends on the molecular architecture
(b) it is a precise breakdown of the material over time
(c) metals biodegrade faster than polymeric materials
(d) all of the above
7. Which class of biomaterials has chemical structures similar to bone?
(a) polymeric biomaterials
(b) ceramic biomaterials
(c) metallic biomaterials
(d) all of the above
8. Which class of biomaterials encourages bonding with surrounding tissues and stimulates new bone growth?
(a) bioinert ceramics
(b) bioactive ceramics
(c) Co-Cr alloys
(d) all of the above
9. Which of the following is TRUE?
(a) ceramics possess excellent wear and friction properties
(b) SS316, Co-Cr and Ti alloys form a protective oxide layer on their surfaces
(c) bioceramics are more reactive then certain metallic implants
(d) all of the above

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